

MUHAMMAD BILAL IRFAN

Associate Software Engineer | Backend & Web Development

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PROFESSIONAL SUMMARY

Computer Science graduate with internship and project experience developing, maintaining, and enhancing web applications using Python, Django REST Framework, REST APIs, PostgreSQL, and SQL. Writes clean, maintainable, object-oriented code, and analyzes, debugs, and resolves software defects down to root cause. Works with Git in an Agile/Scrum environment alongside engineers, QA, and product stakeholders, and uses AI-powered development tools responsibly by validating AI-generated code through critical analysis and testing. Strong analytical and problem-solving ability with working proficiency in written and verbal English; eager to grow as an Associate Software Engineer building healthcare software that improves patient care.

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML, CSS

Frameworks & Libraries: Django, Django REST Framework, LangChain, LangGraph

Databases: PostgreSQL, SQL (queries, joins, schema design, data validation), Database Fundamentals

Web & API Development: REST API design and integration, Web Development, Postman, Stripe payment integration

Core Concepts: Object-Oriented Programming (OOP), Data Structures, Software Design, SDLC, STLC

Development Tools: Git, GitHub, Visual Studio Code, Jira, Postman

Software Quality & Testing: Code Reviews, Debugging, Root Cause Analysis, Test Case Design, Functional Testing, Regression Testing, Smoke Testing, Sanity Testing, UAT, Bug Reporting & Bug Life Cycle

Methodologies: Agile, Scrum, Sprint Ceremonies, Cross-functional Collaboration

AI & Emerging Technologies: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI-assisted development (Claude, ChatGPT, Gemini)

Soft Skills: Analytical Thinking, Problem Solving, Attention to Detail, Teamwork, Communication, Time Management

Languages: English (professional working proficiency - verbal and written), Urdu (native)

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Techtics.AI, Lahore, Pakistan | 2024 - 2025

- Developed, maintained, and enhanced backend modules and product features using Python and Django REST Framework, following established design and coding standards under the guidance of senior engineers.
- Wrote and executed SQL queries against PostgreSQL databases to validate data integrity, confirm expected results, and investigate reported discrepancies.
- Analyzed, debugged, and troubleshooted software defects, identifying root causes, designing test cases and scenarios, logging defects in Jira, and verifying fixes before release to support application stability.
- Built and tested web pages using HTML and CSS, and applied object-oriented programming concepts while implementing, reviewing, and refactoring code modules for readability and maintainability.
- Used Git for version control and branching, and participated in code reviews, technical discussions, and stand-ups, communicating progress and blockers to support sprint objectives.

PROJECTS

MasterwithAwais - Online Course & E-Commerce Platform | masterwithawais.com

Python, Django REST Framework, PostgreSQL, Stripe, HTML, CSS, Git

- Built the backend for an e-commerce learning platform, implementing user authentication and login, course management, and content access control.
- Integrated the Stripe payment gateway to process transactions end to end, handling success, failure, and edge-case flows.
- Designed and queried relational tables for user, course, and order data, then tested login, course access, and payment flows, logging and resolving defects prior to deployment.

FirmFox - Legal Services Platform | dev.firmfox.com

Python, Django REST Framework, LLMs, RAG, LangChain, PostgreSQL

- Contributed to backend development for a legal services web platform, building and maintaining REST API endpoints.
- Implemented and validated LLM and RAG-based features, evaluating outputs for accuracy, and collaborated with the team to troubleshoot issues, weigh technical trade-offs, and deliver features on schedule.

EDUCATION

Bachelor of Science in Computer Science | Pak AIMS - The Institute of Management Sciences, Lahore | 2022 - 2026

Relevant Coursework: OOP, Data Structures & Algorithms, Database Systems, Web Development, Software Engineering.